

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 3B- Introduction to Probability Distributions 2

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Session Outline

1 Cumulative Distribution Functions

2 Moments of Random Variables

- Mathematical Expectation
- Variance

3 Applications

Session Overview

In the previous session, an introduction to probability distributions of discrete and continuous random variables were presented and discussed. This session continues to discuss and introduces students to cumulative distribution functions of random variables. Calculation of probabilities and moments (expectation and variance) using probability distribution functions.

Learning Outcomes

- At the end of the session, students will be able to:
 - 1 Define and Calculate cumulative probability distribution functions for discrete and continuous random variables.
 - 2 Calculate the probabilities of random variables using probability distribution functions.
 - 3 Compute expected and variance of a random variable using probability distribution functions.
 - 4 Sketch the cumulative probability distribution functions graphs for both discrete and continuous random variables.

Cumulative Distribution Function

- The *cumulative distribution function*, or briefly the *distribution function*, for a random variable X is defined by

$$F(x) = P(X \leq x), \quad \text{where } x \text{ is any real number, } (-\infty < x < \infty)$$

- The distribution function $F(x)$ has the following properties:

- 1 $F(x)$ is nondecreasing, that is: $F(x) \leq F(y)$, if $x \leq y$.
- 2 $\lim_{x \rightarrow -\infty} F(x) = 0$; and $\lim_{x \rightarrow \infty} F(x) = 1$.
- 3 $F(x)$ is continuous from the right; that is, $F(x + h)_{h \rightarrow 0^+} = F(x)$ for all values of x .

Distribution Functions for Discrete Random Variables

- The distribution function for a discrete random variable X can be obtained from its probability function by noting that, for all x in, $(-\infty, \infty)$.

$$F(x) = P(X \leq x) = \sum_{u \leq x} P(X = u) = \sum_{u \leq x} p(u)$$

where the sum is taken over all values u taken on by X for which $u \leq x$.

- If the random variable X takes on only a finite number of values $x_1, x_2, \dots, x_n$, then the distribution function is given by

$$F(x) = \begin{cases} 0, & -\infty < X < x_1 \\ p(x_1), & x_1 \leq X < x_2 \\ p(x_1) + p(x_2), & x_2 \leq X < x_3 \\ p(x_1) + p(x_2) + p(x_3), & x_3 \leq X < x_4 \\ \vdots & \vdots \\ p(x_1) + p(x_2) + p(x_3) + \cdots + p(x_n) = 1, & x_n \leq X < \infty \end{cases}$$

Example 1

The probability function corresponding to the random variable X , the number of tails observed for tossing two fair coins is displayed in the table below.

x	0	1	2
$P(X = x)$	0.25	0.5	0.25

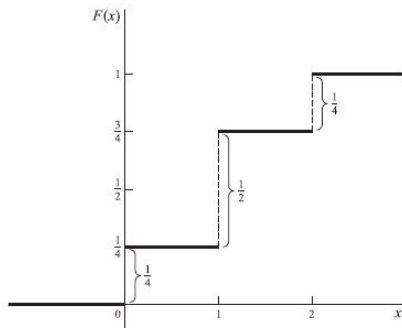
- a Find the distribution function for the random variable X .
- b Obtain its graph.

Solution

- a The distribution function is

$$F(x) = \begin{cases} 0, & -\infty < x < 0 \\ 0.25, & 0 \leq x < 1 \\ 0.75, & 1 \leq x < 2 \\ 1, & 2 \leq x < \infty \end{cases}$$

- b The graph of the distribution function is



Remarks about Discrete D.F.

The following properties about the Example 1, distribution function (df), which are true in general, should be noted.

- 1 The magnitudes of the jumps at 0, 1, 2 are $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$, which are precisely the probabilities. This fact enables one to obtain the probability function from the distribution function.
- 2 Because of the appearance of the graph of Example 1, it is often called a staircase function or step function. The value of the function at an integer is obtained from the higher step; thus the value at 1 is $\frac{3}{4}$ and not $\frac{1}{4}$. This is expressed mathematically by stating that the distribution function is continuous from the right at 0, 1, 2.
- 3 As we proceed from left to right (i.e. going upstairs), the distribution function either remains the same or increases, taking on values from 0 to 1. Because of this, it is said to be a monotonically increasing function.

Continuous Distribution Function

- A nondiscrete random variable X is said to be absolutely continuous, or simply continuous, if its distribution function may be represented as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt, \quad (-\infty < x < \infty)$$

where the function $f(x)$ has the properties

- a $f(x) \geq 0$, for all values of x
 - b $\int_{-\infty}^{\infty} f(x)dx = 1$
- It follows from the above that if X is a continuous random variable, then the probability that X takes on any one particular value is zero, whereas the interval probability that X lies between two different values, a and b , is given by

$$P(a < X < b) = \int_a^b f(x)dx = F(b) - F(a).$$

Example 2

- a Find the constant c such that the function

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0, & \text{Otherwise} \end{cases}$$

is a density function, and

- b Compute $P(1 < X < 2)$.

- a Since $f(x)$ satisfies property 1, if $c \geq 0$, it must satisfy Property 2 in order to be a density function. Now

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^3 cx^2 dx = \left. \frac{cx^3}{3} \right|_0^3 = 9c$$

and since this must equal 1, we have $c = \frac{1}{9}$.

- b Now $f(x) = \frac{1}{9}x^2$ and therefore, we have

$$P(1 < x < 2) = \int_1^2 \frac{1}{9}x^2 dx = \left. \frac{x^3}{9 \times 3} \right|_1^2 = \frac{7}{27}$$

Relationship between PDF and CDF

Let X be a continuous random variable with pdf f and cdf F

- By definition, the cdf is found by integrating the pdf:

$$F(x) = \int_{-\infty}^x f(t) dt$$

- By the Fundamental Theorem of Calculus, the pdf can be found by differentiating the cdf:

$$f(x) = \frac{d}{dx} [F(x)]$$

Example 2

Let the random variable X denote the time a person waits for an elevator to arrive. Suppose the longest one would need to wait for the elevator is 2 minutes, so that the possible values of X (in minutes) are given by the interval $[0,2]$. A possible pdf for X is given by

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2 - x, & 1 \leq x < 2 \\ 0, & \text{otherwise} \end{cases}$$

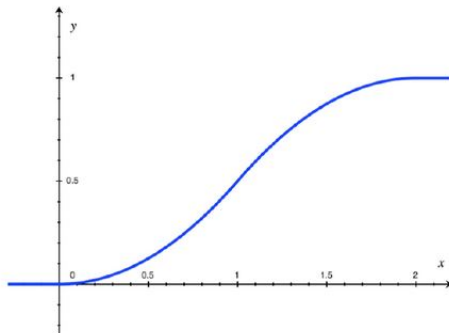
- a Compute the distribution function for $f(x)$.
- b Sketch the graph of $F(x)$.

Solution

- a The distribution function $F(x)$ is given by

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{2}, & 0 \leq x < 1 \\ 2x - \frac{x^2}{2} - 1, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

- b The graph of $F(x)$



Computation of Probabilities

- 1 The probability mass function (p.m.f) of a random variable X is given by

$$P(X = x) = \frac{c\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

where c is a constant.

- a Show that $c = e^{-\lambda}$
- b Find the $P(X = 0)$ and $P(X \geq 2)$.

Solution 1

- a The p.m.f $p(x)$ satisfies property 1 if $c \geq 0$, and must satisfy property 2 if $\sum_{x=0}^{\infty} p(x) = 1$.

$$\Rightarrow \sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{c\lambda^x}{x!} = c \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = ce^{\lambda}$$

$$\therefore c = e^{-\lambda}$$

- b Now

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\therefore P(X = 0) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda}$$

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - [p(0) + p(1)] = 1 - e^{-\lambda} - \lambda e^{-\lambda}.$$

Question 2

Suppose that the time to failure of an electronic component in a copier (in hours) is X with corresponding density function

$$f(x) = \frac{e^{-x/500}}{500}, \quad x > 0.$$

Determine the probability that

- a A component last more than 2000 hours before failure.
- b A component fails in the region from 500 to 2000 hours.
- c A component fails before 500 hours.

Solution 2

- a The probability that a component last more than 2000 hours before failure is:

$$P(X > 2000) = \int_{2000}^{\infty} \frac{e^{-x/500}}{500} dx = -e^{-x/500} \Big|_{2000}^{\infty} = e^{-4}.$$

- b The probability that a component fails in the region from 500 to 2000 hours is:

$$P(500 < X < 2000) = \int_{500}^{2000} \frac{e^{-x/500}}{500} dx = -e^{-x/500} \Big|_{500}^{2000} = e^{-1} - e^{-4}.$$

Solution 2

- The probability that a component fails before 500 hours is

$$P(X < 500) = \int_0^{500} \frac{e^{-x/500}}{500} dx = -e^{-x/500} \Big|_0^{500} = 1 - e^{-1}.$$

Definition of Mathematical Expectation

- A very important concept in probability and statistics is that of the mathematical expectation, expected value, or briefly the expectation, of a random variable.
- For a discrete random variable X having the possible values $x_1, x_2, \dots, x_n$, the expectation of X is defined as

$$\begin{aligned} E(X) &= x_1 P(X = x_1) + x_2 P(X = x_2) + \cdots + x_n P(X = x_n) \\ &= \sum_{i=1}^n x_i P(X = x_i) \end{aligned}$$

- or equivalently, if $P(X = x_i) = p(x_i)$

$$E(X) = x_1 p(x_1) + x_2 p(x_2) + \cdots + x_n p(x_n) = \sum_{i=1}^n x_i p(x_i)$$

- As a special case, where the probabilities are all equal, we have

$$E(X) = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

which is called the *arithmetic mean*, or simply the *mean*, of $x_1, x_2, \dots, x_n$.

- If X takes on an infinite number of values, $x_1, x_2, \dots$, then

$$E(X) = \sum_{i=1}^{\infty} x_i p(x_i),$$

provided that the infinite series converges absolutely.

Mathematical Expectation

- For a continuous random variable X having density function $f(x)$, the expectation of X is defined as

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx,$$

provided that the integral converges absolutely.

- The expectation of X is very often called the mean of X and is denoted by μ_x , or simply μ , when the particular random variable is understood.
- The mean, or expectation, of X gives a single value that acts as a representative or average of the values of X , and for this reason it is often called a *measure of central tendency*.

Example 3

Suppose that a game is to be played with a single die assumed fair. In this game a player wins GHs 20 if a 2 turns up, GHs 40 if a 4 turns up; loses GHs 30 if a 6 turns up; while the player neither wins nor loses if any other face turns up. Find the expected sum of money to be won.

Solution:

Let X be the random variable giving the amount of money won on any toss. The possible amounts won when the die turns up 1, 2, 3, 4, 5, 6 are $x_1, x_2, \dots, x_6$, respectively.

The expected value or expectation is

$$E(X) = 0\left(\frac{1}{6}\right) + 20\left(\frac{1}{6}\right) + 0\left(\frac{1}{6}\right) + 40\left(\frac{1}{6}\right) + 0\left(\frac{1}{6}\right) - 30\left(\frac{1}{6}\right) = 5$$

X	0	+20	0	+40	0	-30
$p(x_i)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Example 4

The density function of a random variable X is given by

$$f(x) = \begin{cases} \frac{1}{2}x, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute the expected value of X .

Solution:

The expected value of X is

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^2 x \left(\frac{1}{2}x \right) dx = \int_0^2 \frac{x^2}{2} dx \\ &= \frac{x^3}{6} \Big|_0^2 = \frac{4}{3} \end{aligned}$$

Some Properties of Expectation

- 1 If $c \neq 0$ is any constant, then

$$E(cX) = cE(X)$$

Note: If $b > 0$ is any constant, then

$$E(cX + b) = cE(X) + b$$

- 2 If X and Y are any random variables, then

$$E(X + Y) = E(X) + E(Y)$$

- 3 If X and Y are independent random variables, then

$$E(XY) = E(X)E(Y)$$

Variance and Standard Deviation

- 1 Another quantity of great importance in probability and statistics is called the variance and is defined by

$$\text{Var}(X) = E[(X - \mu)^2]$$

- 2 The variance is a nonnegative number. The positive square root of the variance is called the standard deviation and is given by

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{E[(X - \mu)^2]}$$

- 3 Where no confusion can result, the standard deviation is often denoted by σ instead of σ_X , and the variance in such case is σ^2 .

Variance and Standard Deviation

- i If X is a discrete random variable taking the values $x_1, x_2, \dots, x_n$ and having probability function $p(x_i)$ then the variance is given by

$$\text{Var}(X) = E[(x - \mu)^2] = \sum_{i=1}^n (x_i - \mu)^2 p(x_i)$$

- ii In the special case where the probabilities are all equal, we have

$$\text{Var}(X) = E[(x - \mu)^2] = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_n - \mu)^2}{n}$$

which is the variance for a set of n numbers $x_1, x_2, \dots, x_n$.

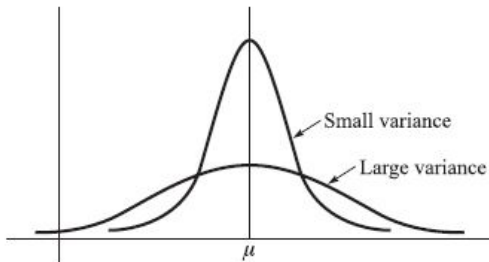
- iii If X is a continuous random variable having density function $f(x)$, then the variance is given by

$$\text{Var}(X) = E[(x - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

provided that the integral converges.

Variance and Standard Deviation

- The variance (or the standard deviation) is a measure of the dispersion, or scatter, of the values of the random variable about the mean .
- If the values tend to be concentrated near the mean, the variance is small;
- while if the values tend to be distributed far from the mean, the variance is large.
- A graphical case of two continuous distributions having the same mean, μ .



Example 5

Find the variance and standard deviation of the random variable X , having the density function $f(x)$ with mean $\mu = 4/3$.

$$f(x) = \begin{cases} \frac{1}{2}x, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Solution:

Given that $\mu = 4/3$ we have Variance given by

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] = \int_0^2 \left(X - \frac{4}{3}\right)^2 \left(\frac{1}{2}x\right) dx \\ &= \int_0^2 \left(x^2 - \frac{8}{3}x + \frac{16}{9}\right) \left(\frac{1}{2}x\right) dx = \frac{x^4}{8} - \frac{4}{9}x^3 + \frac{4}{9}x^2 \Big|_0^2 \\ &= 2 - \frac{32}{9} + \frac{16}{9} = \frac{2}{9} \quad \text{and} \quad \sigma = \frac{\sqrt{2}}{3} \square \end{aligned}$$

Some Theorems of Variance

1 Theorem 1:

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - \mu^2$$

where $\mu = E(X)$.

2 Theorem 2: If c and b are any constants,

$$\text{Var}(cX + b) = c^2 \text{Var}(X)$$

3 Theorem 3: If X and Y are independent random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$$

4 Theorem 4: If X and Y are not independent random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

Solved Examples

Question 1: In a lottery there are 200 prizes of \$5, 20 prizes of \$25, and 5 prizes of \$100. Assuming that 10,000 tickets are to be issued and sold, what is a fair price to pay for a ticket? *Solution:*

Let X be a random variable denoting the amount of money to be won on a ticket. The various values of X together with their probabilities are shown in the table below:

X in Dollars	5	25	100	0
$P(X = x)$	0.02	0.002	0.0005	0.9775

The expectation of X in dollars is thus

$$E(X) = \sum xp(x) = 5(0.02) + 25(0.002) + 100(0.0005) + 0(0.9775) = 0.2$$

Thus the fair price to pay for a ticket is 20 cents.

Solved Examples Cont'd

Question 2: Find the expectation of the sum of points in tossing a pair of fair dice.

Solution:

Let X and Y be the points showing on the two dice. We have

$$E(X) = E(Y) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \cdots + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

$$\therefore E(X + Y) = E(X) + E(Y) = 7$$

Solved Examples Cont'd

Question 3: A continuous random variable X has probability density given by

$$f(x) = \begin{cases} 2e^{-2x}, & 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

Find

- a Expectation of X , $E(X)$
- b $E(X^2)$
- c $Var(X)$

Solved Examples Cont'd

Solution: (a)

$$\begin{aligned} E(X) &= \int_0^{\infty} xf(x)dx = \int_0^{\infty} x(2e^{-2x})dx = 2 \int_0^{\infty} xe^{-2x}dx \\ &= 2\left(\frac{1}{4}\right) = \frac{1}{2}. \end{aligned}$$

The integration above is a standard integral so by using the gamma function formula, that is:

$$\int_0^{\infty} x^{r-1} e^{-\beta x} dx = \frac{(r-1)!}{\beta^r}$$

Solved Examples Cont'd

Solution: (b)

$$\begin{aligned} E(X^2) &= \int_0^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 (2e^{-2x}) dx = 2 \int_0^{\infty} x^2 e^{-2x} dx \\ &= 2 \left(\frac{2!}{2^3} \right) = \frac{1}{2}. \end{aligned}$$

(c): The variance is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \quad \square$$

THANK YOU!!!